

PART 3 DEEP DIVE

AI Terms for Beginners

*Tokens, Embeddings, Temperature,
Few-shot, Inference & Guardrails*

in-depth explanations, real examples,
flowcharts + a beginner project for each

WHAT IT MEANS

Tokens are the small pieces AI breaks your text into before it can read or write anything. It doesn't see full sentences like we do — it sees a string of small chunks (often close to words or parts of words), one after another.

REAL-LIFE EXAMPLE

When you type "I love AI" into ChatGPT or Claude, it doesn't read that as one sentence. It splits it into pieces like "I", "love", "AI" — similar to how Instagram counts characters in your caption, except AI counts and processes tokens instead.

HOW IT WORKS



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Build a simple "Token Counter" using Python. Take any sentence as input, split it into words, and print how many tokens (words) it has. This teaches you exactly what AI "sees" before it responds.

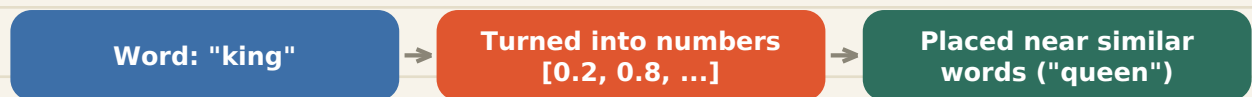
WHAT IT MEANS

Embeddings turn words, sentences, or images into numbers (called vectors) so AI can measure how similar two things are. Words with similar meaning end up with similar numbers, even if the actual words are totally different.

REAL-LIFE EXAMPLE

This is exactly how Spotify's "Recommended for you" or YouTube's "Up next" works. It doesn't understand music or video the way you do — it converts each song/video into numbers and finds others whose numbers are close by.

HOW IT WORKS



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Use a free embeddings tool (like OpenAI's or a Python library such as sentence-transformers) to compare two sentences and print a "similarity score" between them. Try similar and very different sentences and see the score change.

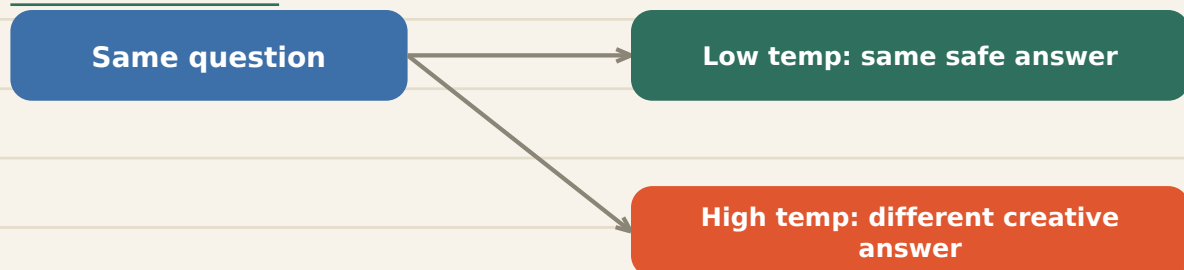
WHAT IT MEANS

Temperature is a setting that controls how random or creative an AI's answers are. Low temperature = safe, predictable, same-ish answers every time. High temperature = more variety and creativity, but sometimes less accurate.

REAL-LIFE EXAMPLE

Think of Google Maps vs. a well-travelled friend. Maps (low temperature) gives you the same optimal route every time. Your friend (high temperature) might suggest a different, more interesting route each time you ask — sometimes great, sometimes not ideal.

HOW IT WORKS



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If you're using an AI API (like OpenAI or Claude), send the exact same prompt twice — once with a low temperature (like 0.1) and once with a high temperature (like 1.0). Compare how different the two answers are.

WHAT IT MEANS

Zero-shot means asking AI to do a task with zero examples — it just uses what it already knows. Few-shot means giving it 2-3 examples first, so it can copy the pattern before answering your real question.

REAL-LIFE EXAMPLE

It's like a classroom test. Zero-shot is answering a question cold, with no prep. Few-shot is like a teacher showing you 2-3 solved examples on the board before you attempt a similar problem — you instantly understand the expected pattern.

HOW IT WORKS



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Pick a small task, like sorting messages into "Urgent" or "Not Urgent." First ask AI to do it with zero examples. Then give it 3 solved examples and ask again. Compare which one gives better results.

WHAT IT MEANS

Inference is the moment AI actually uses everything it learned during training to answer your specific question. It's not learning anything new in that moment — it's just applying what it already knows, instantly.

REAL-LIFE EXAMPLE

Like Face ID unlocking your phone. It doesn't relearn your face every time you unlock — it already learned it once, and now it just quickly checks ("infers") if this new face matches. That quick check is inference.

HOW IT WORKS



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If you use any pre-trained model (like a sentiment analysis model from HuggingFace), run 5 different sentences through it and note the outputs. You're doing "inference" — using a trained model without training it yourself.

WHAT IT MEANS

Guardrails are safety rules built into AI so it doesn't give harmful, biased, or completely off-topic answers. They act like a filter that checks a response before or after it's generated.

REAL-LIFE EXAMPLE

Similar to YouTube's Restricted Mode or Kids profile — content is automatically filtered before it reaches the viewer. AI guardrails work the same way: filtering unsafe or inappropriate responses before you ever see them.

HOW IT WORKS

User asks something



Safety rules check it



Safe answer or polite
refusal

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Write a simple Python function that acts as a mini "guardrail": if user input contains certain blocked words, print "Sorry, I can't help with that." Otherwise, pass it along normally. This is a simplified version of how real guardrails work.